

Hexstrike-AI vs Gemini-CLI: AI-Powered Offensive Security Tools on Kali Linux

Author: Arnel C. Reyes

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Artificial intelligence is redefining offensive security, and Hexstrike-AI and Gemini-CLI are at the forefront of this transformation. Both tools leverage AI to enhance penetration testing workflows on Kali Linux, but they do so in distinct ways. Hexstrike-AI focuses on exploit generation and post-exploitation automation, streamlining complex attack chains through intelligent orchestration. Gemini-CLI, meanwhile, integrates large language model capabilities directly into the terminal, enabling AI-assisted reconnaissance, payload crafting, and adaptive attack strategies.

Together, these tools represent the next evolution of the red-team and penetration testing landscape—where human expertise is amplified by machine intelligence, and the boundaries of automation are continually expanding.

Background

The offensive side of cybersecurity is undergoing a shift. Traditionally, penetration testing relied heavily on manual enumeration, scripting and deep domain knowledge. With systems growing more complex and threat surfaces expanding, there's an increasing need for automation that isn't just "run this script" — but "intelligently adapt to what the system gives you."

In this context:

- Hexstrike-AI brings in neural- or model-based steps to autonomously pick exploit paths, deliver payloads and escalate privileges.
- Gemini-CLI takes a different tack, using AI to assist in reconnaissance, make sense of system responses, refine attack vectors and craft more effective payloads.

Both tools represent a significant shift in how penetration testing is performed. Instead of static checklists or prebuilt scripts, AI is now bringing context-aware adaptability to the attacker's toolkit. Kali Linux provides the ideal sandbox — flexible, open, and battle-tested — for experimenting with these next-generation offensive systems. Running on Kali Linux gives penetration testers and red-teams a familiar environment to test these next-gen tools.

Preparing Kali Linux

Kali Linux is a powerful Debian-based penetration testing distribution packed with security tools. It can be ran on a Virtual Machine (VM) or in a Docker container.

Option A: Running Kali Linux on a Virtual Machine

- Download the official Kali Linux image

Go to the official site: <https://www.kali.org/get-kali/>

- VMware Image (.ova)
- VirtualBox Image (.ova)
- ISO (for manual installation)

- Import Kali into your virtualization software

- For VirtualBox:
 - Open VirtualBox → **File** → **Import Appliance**
 - Select the **.ova** file
 - Click **Import**
- For VMWare:
 - Open VMware Workstation or Fusion → **File** → **Open**
 - Select the **.ova** file
 - Click **Import**

- Start Kali Linux

Default credentials (from official images):

Username: **kali**
Password: **kali**

- After login, update everything:

```
$ sudo apt update && sudo apt full-upgrade -y
```

- (Optional) Install full suite of tools

```
$ sudo apt install -y kali-linux-full
```

> Installs *all* tools.
> If preferred just the top tools:

```
$ sudo apt install -y kali-linux-top10
```

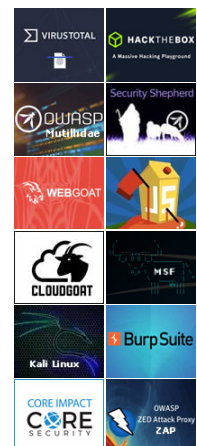
kali-linux-top10 is a metapackage that pulls in the 10 most important Kali tools. Installing it will install these packages:

- aircrack-ng — wireless LAN (WEP/WPA) cracking suite
- burpsuite — web application testing proxy / scanner (community or bundled edition)
- hydra — fast network login cracker (password brute force)
- john — John the Ripper, password-cracking suite
- metasploit-framework — exploitation framework (msfconsole, modules, etc.)

Application Security



Ethical Hacking



- netexec — small remote command execution helper (used in some pentest workflows)
- nmap — network scanner / port scanner / host discovery
- responder — LLMNR/NBT-NS/mDNS poisoning and credential capture tool
- sqlmap — automated SQL injection and database takeover tool
- wireshark — network protocol analyzer / packet capture GUI

Option B: Running Kali Linux on a Docker Container

1. Pull the official Kali Linux image

```
$ docker pull kalilinux/kali-rolling
```

2. Run an interactive Kali container

```
$ docker run -it --name mykali kalilinux/kali-rolling /bin/bash
```

Expose ports if needed (for Hexstrike-AI MCP Server)

Example: Expose Hexstrike-AI on port 8888

```
$ docker run -it -p 8888:8888 --name mykali kalilinux/kali-rolling /bin/bash
```

3. Update packages

```
$ apt update && apt full-upgrade -y
```

4. Install essential tools

For top tools:

```
$ apt install -y kali-linux-top10
```

Or full suite:

```
$ apt install -y kali-linux-full
```

Or everything

```
$ apt install -y kali-linux-everything
```

Warning: This will take time and disk space (20–40 GB).

5. Add networking tools

```
$ apt install -y net-tools iproute2 traceroute dnsutils ethtool wireless-tools arp-scan curl wget
```

6. Install pip3 if needed

```
$ apt install -y python3-pip
```

7. Save and reuse by keeping container persistent

To keep tools and data after exiting by committing a container to an image

```
$ docker commit mykali mykali-custom
```

Next time, just run:

```
$ docker run -it --name kali mykali-custom /bin/bash
```

8. If preferred a custom Kali image with preinstalled tools:

Create Dockerfile file:

```
``Dockerfile
FROM kalilinux/kali-rolling

RUN apt update && apt install -y \
    nmap \
    metasploit-framework \
    sqlmap \
    hydra \
    nikto \
    net-tools \
    iputils-ping \
    iproute2 \
    traceroute \
    dnsutils \
    ethtool \
    wireless-tools \
    arp-scan \
    gemini-cli \
    && apt clean
``
```

Then build it:

```
$ docker build -t mykali .
```

Then run it:

```
$ docker run -it mykali
```

Next time, run using the following:

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```
$ docker start -i mykali
```

Interact with running container:

```
$ docker exec -it mykali /bin/bash
```

Hexstrike-AI Installation

Hexstrike-AI Proof of Concept

Gemini-CLI Installation

Gemini-CLI Proof of Concept

Hexstrike-AI is purpose-built to automate offensive workflows end-to-end. It offers a multi-agent architecture, tying large language models (LLMs) like GPT or Claude into a decision-engine that orchestrates over 150 professional security tools (scanning, exploitation, payload delivery, binary & web apps) in a unified framework. Because of this high level of automation, the tool is already attracting attention — both from red teams and, unfortunately, threat actors. For example, it reportedly reduced the time-to-exploit for certain CVEs from days to under 10 minutes.

Setting-Up Hexstrike-AI on Kali

1. Clone the repository

```
$ git clone https://github.com/0x4m4/hexstrike-ai.git
$ cd hexstrike-ai
```

2. Setup virtual environment

Install virtual environment

```
$ apt -y install python3.13-venv
```

Create virtual environment:

```
$ python3 -m venv hexstrike-env
$ source hexstrike-env/bin/activate # Linux/Mac
# hexstrike-env\Scripts\activate # Windows
```

3. Install Python dependencies

```
$ apt install -y python3-pip
$ pip3 install -r requirements.txt
```

4. Start the backend Hexstrike-AI agent server

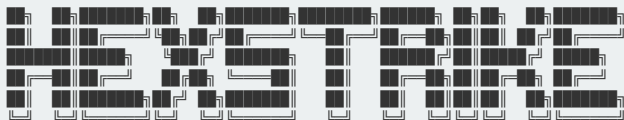
```
$ python3 hexstrike_server.py

# Optional: Start with debug mode
$ python3 hexstrike_server.py --debug

# Optional: Custom port configuration
$ python3 hexstrike_server.py --port 8888
```

If backend agent server is successfully setup, the server will start listening:

```
2025-10-22 13:38:55,417 - __main__ - INFO - 🛠 Process pool worker 0 started
2025-10-22 13:38:55,419 - __main__ - INFO - 🛠 Process pool worker 1 started
2025-10-22 13:38:55,420 - __main__ - INFO - 🛠 Process pool worker 2 started
2025-10-22 13:38:55,421 - __main__ - INFO - 🛠 Process pool worker 3 started
```



```
🚀 HexStrike AI - Blood-Red Offensive Intelligence Core
⚡ AI-Automated Recon | Exploitation | Analysis Pipeline
🕒 Bug Bounty | CTF | Red Team | Zero-Day Research
```

```
[INFO] Server starting on 127.0.0.1:8888
[INFO] 150+ integrated modules | Adaptive AI decision engine active
[INFO] Blood-red theme engaged - unified offensive operations UI
```

```
2025-10-22 13:38:55,499 - __main__ - INFO -
2025-10-22 13:38:55,499 - __main__ - INFO -
2025-10-22 13:38:55,500 - __main__ - INFO -
2025-10-22 13:38:55,500 - __main__ - INFO -
2025-10-22 13:38:55,500 - __main__ - INFO -
2025-10-22 13:38:55,501 - __main__ - INFO -
2025-10-22 13:38:55,501 - __main__ - INFO -
2025-10-22 13:38:55,501 - __main__ - INFO -
2025-10-22 13:38:55,501 - __main__ - INFO -
2025-10-22 13:38:55,501 - __main__ - INFO -
2025-10-22 13:38:55,501 - __main__ - INFO -
* Serving Flask app 'hexstrike_server'
* Debug mode: off
2025-10-22 13:38:55,515 - werkzeug - INFO - WARNING: This is a development server. Do not
use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:8888
* Running on http://172.17.0.2:8888
2025-10-22 13:38:55,515 - werkzeug - INFO - Press CTRL+C to quit
```

```
🚀 Starting HexStrike AI Tools API Server
🌐 Port: 8888
🛠 Debug Mode: False
📦 Cache Size: 1000 | TTL: 3600s
🕒 Command Timeout: 300s
⚡ Enhanced Visual Engine: Active
```

5. Verify Installation

```
# Test server health
$ curl http://localhost:8888/health
```

```
# Test AI agent capabilities
$ curl -X POST http://localhost:8888/api/intelligence/analyze-target \
-H "Content-Type: application/json" \
-d '{"target": "example.com", "analysis_type": "comprehensive"}'
```

Setting-Up Hexstrike-AI on AI Client

Option A: Claude

1. Download and Install Claude

Go to: <https://claude.com/download>

2. Once installed, login using **Claude** account

To configure:

- Go to **Settings** → **Developer** → **Local MCP Servers**
- Click **Edit Config**
- Edit `~/claude/claude_desktop_config.json`

```
{
  "mcpServers": {
    "hexstrike-ai": {
      "command": "python",
      "args": [
        "/path-to/hexstrike-ai/hexstrike_mcp.py",
        "--server",
        "http://ip-address-of-hexstrike-ai-backend-server:8888"
      ]
    }
  }
}
```

- Reset Claude
- Once Claude is restarted, hexstrike-ai tool must be available on tools list
- AI client is ready to receive prompt to interact with Hexstrike-AI to perform security tests

Option B: Cursor

1. Download and Install Cursor

Go to: <https://cursor.com/download>

2. Once installed, login using **Cursor** account

To configure:

- Open **Settings (Ctrl+Shift+J)** → **Tools & MCP** → **Tools**
- Click **New MCP Server**
- Edit `~/.cursor/mcp.json`

```
{
  "mcpServers": {
    "hexstrike-ai": {
      "command": "python",
      "args": [
        "/path-to/hexstrike-ai/assets/hexstrike_mcp.py",
        "--server",
        "http://ip-address-of-hexstrike-ai-backend-server:8888"
      ],
      "description": "HexStrike AI v6.0 - Advanced Cybersecurity Automation Platform",
      "timeout": 700,
      "disabled": false
    }
  }
}
```

- AI client is ready to receive prompt to interact with Hexstrike-AI to perform security tests

Option C: Sire

1. Download and Install Sire

Go to: <https://Sire.app/>

2. Once installed, login using **Sire** account

To configure:

- Go to **Tools** → **Tools**
- Click + **Local**
- Enter the following:

Tool Key: **hexstrike**
Name: **hexstrike**
Approval Policy: **Never**
Command: **python /path-to/hexstrike-ai/hexstrike_mcp.py --server http://ip-address-of-hexstrike-ai-backend-server:8888**

Config Preview:

```
{
  "name": "hexstrike",
  "key": "hexstrike",
```

```
"description": "hexstrike",
"command": "python",
"args": [
  "/path-to/hexstrike-ai/hexstrike_mcp.py",
  "--server",
  "http://ip-address-of-hexstrike-ai-backend-server:8888"
],
"approvalPolicy": "never"
}
```

- Click **Save**
- AI client is ready to receive prompt to interact with Hexstrike-AI to perform security tests

More details about Hexstrike-AI: <https://github.com/0x4m4/hexstrike-ai>

Comparison: Hexstrike-AI vs. Gemini-CLI

Feature	Hexstrike-AI	Gemini-CLI Native
AI Model Support	Supports any LLM (e.g., Claude, GPT, Copilot, Deepseek) via MCP integration; highly flexible for custom setups.	Limited to Google's Gemini models (e.g., Gemini 2.5 Pro); no support for external LLMs.
Tool Integration	Autonomous execution of 150+ Kali tools (e.g., Nmap, sqlmap, Metasploit, hashcat) across categories like reconnaissance, web exploits, and reverse engineering.	Integrates with Kali's core tools (e.g., Nmap, sqlmap, wfuzz) for chained automation; focuses on OWASP Top 10 and common workflows.
Automation Level	High autonomy with multi-agent architecture; AI agents make decisions, discover attack chains, and run full pentests independently.	Semi-autonomous; natural language prompts trigger steps, but requires human review/confirmation (interactive mode) or "YOLO" mode for full auto.
Use Cases	Advanced pentesting, bug bounty automation, zero-day exploitation, security research; excels in complex, multi-stage scenarios.	Reconnaissance, enumeration, vulnerability scanning, remediation suggestions; best for quick, guided assessments.
Installation	Git clone from GitHub, Python venv setup, pip dependencies; requires manual tool installs (script available); ~50GB+ storage for full setup.	Native via <code>sudo apt install gemini-cli</code> ; lightweight (12MB installed size); no extra dependencies beyond Kali 2025.3.
System Requirements	Kali 2024.1+, Python 3.9+, 8GB+ RAM, 50GB+ storage, GPU optional for acceleration.	Kali 2025.3+, minimal resources; runs efficiently on standard hardware.
Human Oversight	Optional; designed for full autonomy but supports intervention via API/logs.	Emphasizes control; prompts for confirmation on actions to prevent misuse.
Ethical/Safety Features	Logging and caching for audit; creator emphasizes defender use, but adaptable for red teaming.	Built-in consent prompts, operation logging, offline mode; safeguards against over-reliance or abuse.
Community & Updates	Active GitHub repo (0x4m4/hexstrike-ai); v6.0 includes vulnerability intelligence; community-driven tool additions.	Official Kali package; tied to Google's Gemini updates; integrated into Kali's rolling release cycle.

Looking at the Future to Advance Offensive Security Using AI

Hexstrike-AI represents a highly automated, tool-chain-oriented, end-to-end AI framework capable of performing reconnaissance, scanning, exploitation, and persistence with minimal manual intervention. This level of automation introduces significant efficiency gains but also comes with added complexity and potential risk.

Gemini-CLI, on the other hand, serves as an LLM-powered assistant integrated into the penetration testing workflow. It provides a conversational interface for crafting reconnaissance queries, generating command snippets, and assisting in payload development—while maintaining the operator's control over execution.

Together, these tools demonstrate how artificial intelligence is transforming offensive security from traditional "manual toolsets and scripts" into "AI-assisted reasoning and automation." However, they also emphasize an essential principle: technology cannot replace human judgment, authorization, or ethical responsibility. When applied responsibly, AI tools act as powerful force multipliers; when misused, they can amplify risk and unintended consequences.

Conclusion

What this comparison shows is that AI-powered offensive tools are no longer just hype — they're becoming practical. Hexstrike-AI shines when you need a more autonomous exploit-and-escalate chain. Gemini-CLI stands out when you want flexibility, data-driven reconnaissance and payload refinement.

For anyone in offensive security on Kali Linux: these tools represent the next wave — faster, smarter, more adaptive. That said, they don't replace human expertise, judgment or ethics. Used well, they're force multipliers. Used poorly, they can raise serious risk.

In short: Hexstrike-AI and Gemini-CLI signal that offensive security is evolving — and practitioners who stay ahead of the curve will be better equipped..

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